

MARROW

The Living Blood Network

A predictive, agentic and federated AI platform that turns thalassemia care coordination from reactive crisis management into a planned, human-centered system of continuity.

AI FOR BLOOD DONATION & CARE COORDINATION

TEAM FLUX
VASU PARASHAR · HARSH TICKU

— THE PROBLEM

Heroic effort, exhausted systems.

1,00,000+

Thalassemia patients across India

500-700

Lifetime transfusions per patient

Every 2-4 weeks

Recurring, predictable demand - yet still treated as an emergency

WHY TODAY'S SYSTEM BUCKLES

Manual matching at scale

Coordinators rely on spreadsheets and phone calls; cognitive load grows linearly with patient count.

Fragmented coordination

Patients, donors, hospitals and NGOs live on different channels, in different languages.

Engagement drift

Donors silently disengage within months without continuous, personalised reinforcement.

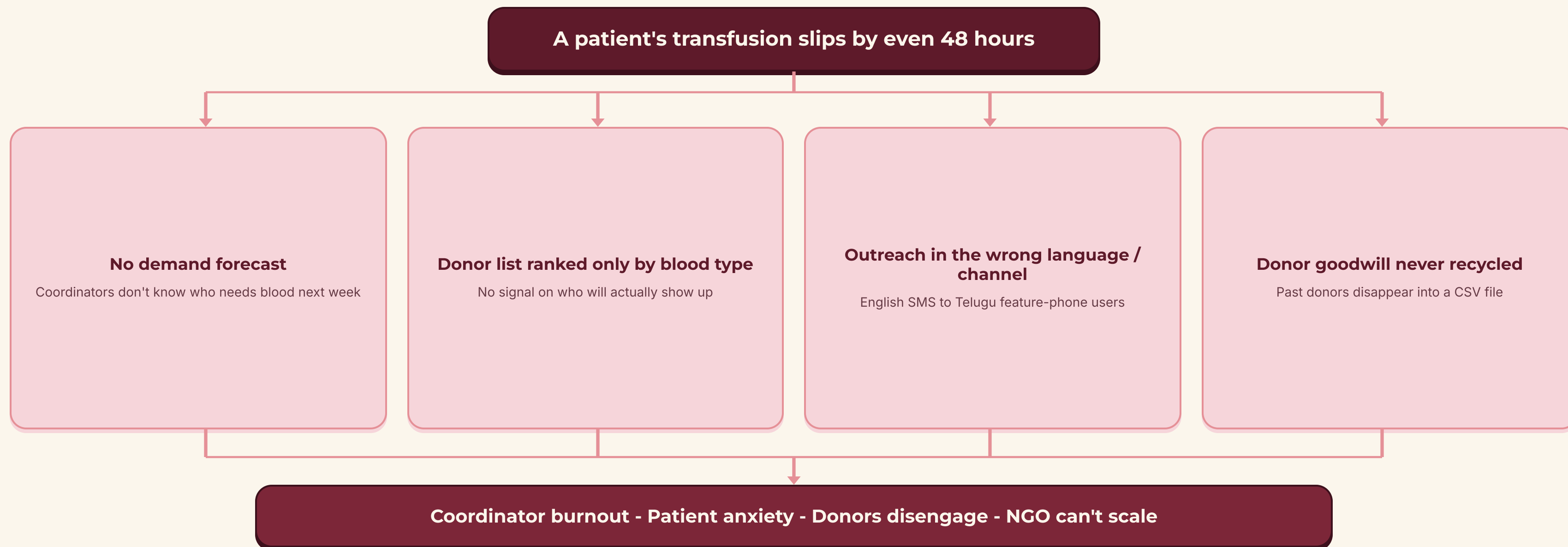
Hard-to-sustain scale

Each new city, hospital and language adds friction the current architecture cannot absorb.

— ANATOMY OF THE PROBLEM

One missed match, a cascade of failures.

Every late transfusion is the visible tip of a deeper chain reaction. Understanding the chain is what lets us interrupt it.



— THE CORE INSIGHT

From reactive panic to predictive planning.

Most blood-donation tools treat every transfusion as a sudden emergency. But thalassemia demand is not chaotic - it is rhythmic, recurring, and forecastable for each patient. Marrow reframes the whole problem.

Today

Pull-based panic

- Patient needs blood → SOS broadcast goes out
- Coordinator scrambles, calls dozens manually
- Donor list ranked only by blood type
- Each cycle starts from a blank slate
- No memory carries across interactions

With Marrow

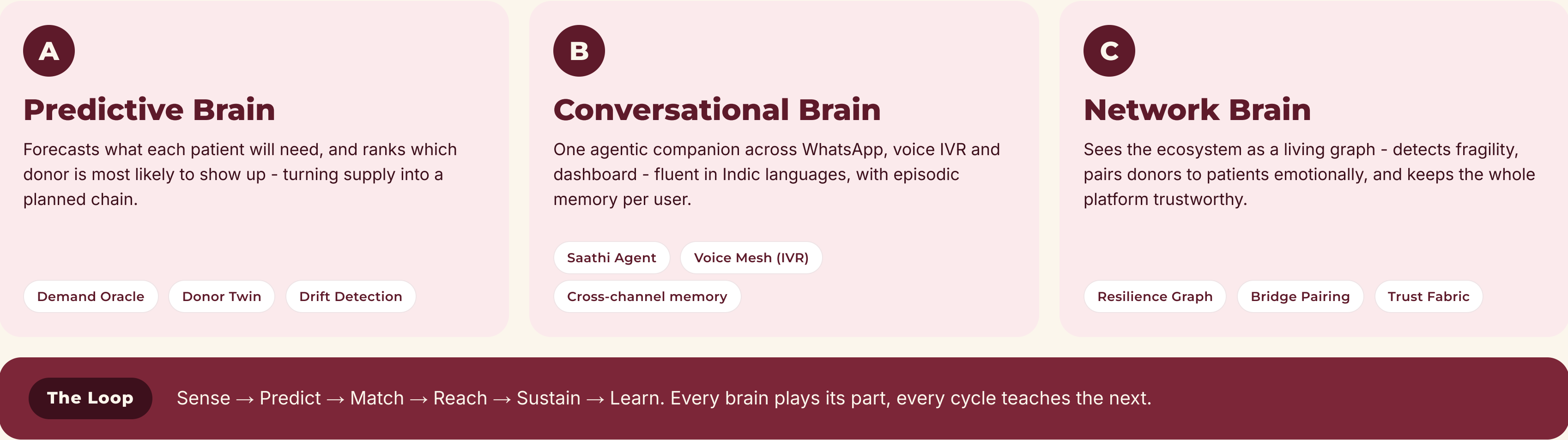
Push-based planning

- Patient demand forecast weeks ahead of time
- Top donors pre-staged by propensity, not blood type alone
- Multi-language outreach, channel-aware, memory-aware
- Engagement learns and adapts per donor
- Network fragility detected proactively

— OUR APPROACH

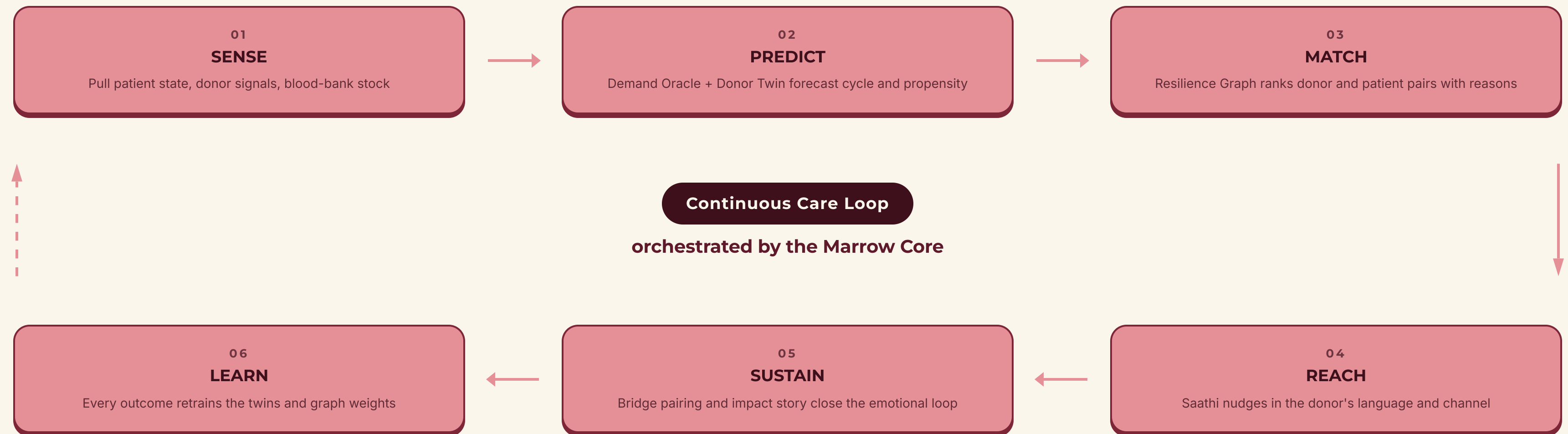
Three brains, one loop, one living network.

Marrow organises itself as three specialised brains - Predictive, Conversational, Network - coordinated by a single orchestrator that drives a continuous care loop across the entire ecosystem.



— THE OPERATING MODEL

A loop, not a pipeline. Every cycle teaches the next.



— BRAIN A • THE PREDICTIVE BRAIN

Forecasting demand, ranking willingness.

Demand Oracle

A digital twin for every patient

Learns each patient's transfusion rhythm - hemoglobin decay, weight, age, splenectomy status - and outputs calibrated forecasts.

OUTPUT

"Aarav will need O+ on Nov 24, ±2 days."

- Per-patient forecast horizon: 14-28 days
- City-level aggregation for hospitals + blood banks
- Converts on-demand panic into a planned supply chain

Prophet

LSTM

Per-patient twin

Donor Twin

A behavioural model per donor

Eligibility clock · response history · geo-temporal availability · channel preference · emotional engagement state. A drift detector flags donors going dormant *before* they fully disengage.

OUTPUT

"Priya · 86% likely · evening · WhatsApp · Telugu."

- XGBoost-based propensity scoring
- Lifecycle state machine: active, drifting, dormant, lapsed
- Re-activation playbooks per state

Propensity

Drift detection

Lifecycle state

— BRAIN B · THE CONVERSATIONAL BRAIN

One agentic companion, every channel, every language.

Saathi · Agentic Companion

Four personas, one shared brain.

A LangGraph-orchestrated agent with episodic memory across sessions, languages and channels. The persona adapts to who is speaking - the tone changes, the brain doesn't.

Patient parent

Gentle reminders, reassurance, medication tracking in their language.

Donor

Personalised nudges, eligibility window, transparent impact updates.

Coordinator

Natural-language queries over the donor + patient graph.

Hospital

Slot confirmations, intake sync, inventory pings via API.

Voice Mesh

Saathi over a phone call.

A voice-first IVR layer for feature-phone users - same brain, delivered by phone call, in native languages via the Bhashini stack.

- Register, confirm, hear impact - all by voice
- Memory shared with WhatsApp + dashboard channels
- Reaches rural India directly, no app required
- Inbound and outbound (proactive nudges)

Bhashini ASR + TTS

Exotel / Twilio

Cross-channel memory

A donor who confirmed by voice yesterday is greeted by name on WhatsApp tomorrow. The memory layer (ChromaDB + structured store) carries context across every touchpoint.

— BRAIN C • THE NETWORK BRAIN

Seeing the network as a living graph.

The Network Brain treats the entire ecosystem - patients, donors, hospitals, blood banks - as a heterogeneous graph. It detects fragility, builds emotional bridges, and keeps the platform trustworthy at scale.

Resilience Graph

Detect collapses before they happen.

- Fragile patient detection (low donor coverage)
- Donor cold-spot mapping by geography
- Bridge-broker identification
- Lookalike donor recruitment

GraphSAGE

NetworkX

Bridge Pairing

From transactions to relationships.

Each donor is paired with a specific patient they are sustaining. Every donation closes the loop with an AI-generated personal update - not a points balance, a story.

Pair → Donate → Witness → Reward → Constellate

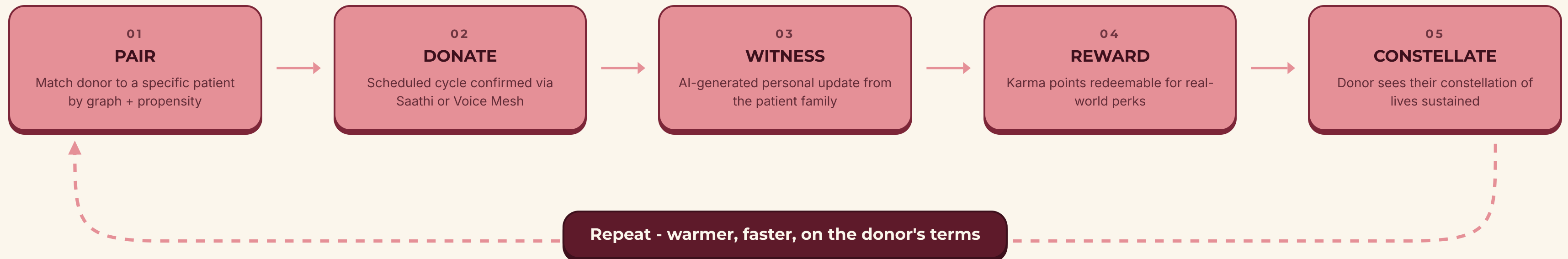
Trust Fabric

Built for the real world.

- Federated learning across hospitals (Flower)
- Differential privacy on aggregate forecasts
- Explainable matching with plain-language reasons
- Consent ledger aligned with India's DPDP Act

— ENGAGEMENT, ENGINEERED

Why donors come back: they are sustaining a person, not a database.



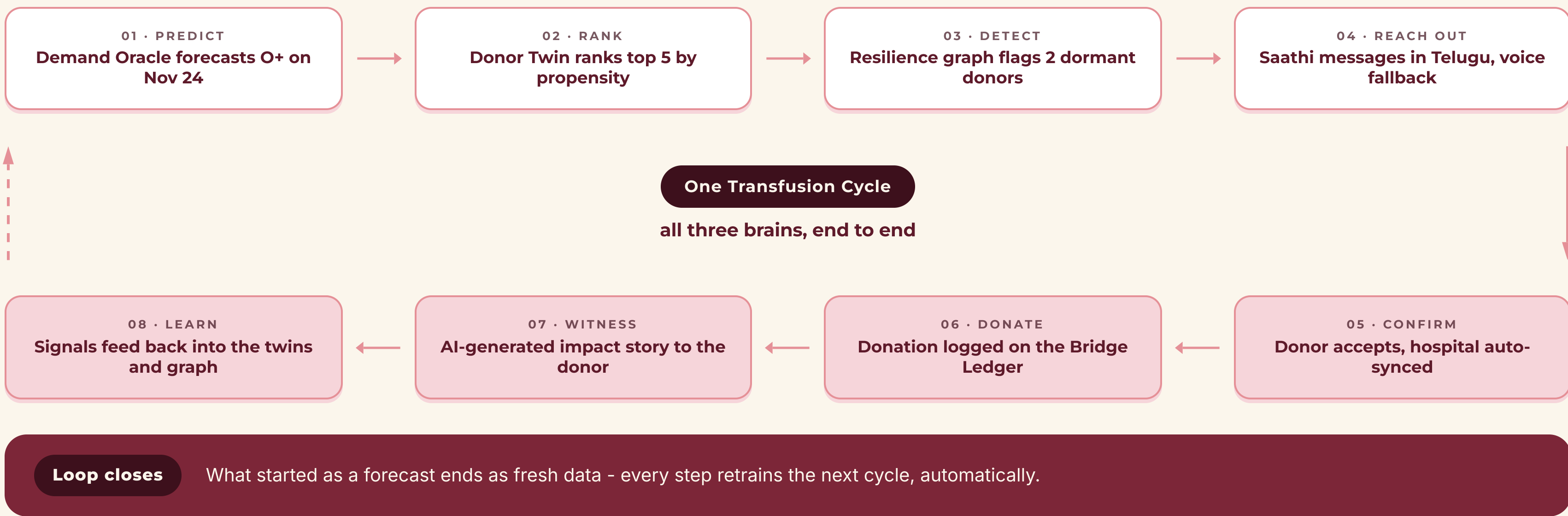
Why it works

A motivation classifier picks the right reward type per donor - gratitude, recognition, social, or tangible - so retention rises without manipulation.

— END-TO-END WALKTHROUGH

Aarav, age 9, Hyderabad.

One cycle, every brain.



— MAPPING TO THE BRIEF

Every requirement, mapped to a brain.

REQUIREMENT FROM THE BRIEF	HOW MARROW SOLVES IT
Stay connected at the right time	Saathi + Demand Oracle proactive triggers
Intuitive cross-language interactions	Saathi multilingual stack + Voice Mesh
Remember across repeated interactions	Episodic memory in the vector store
Identify and prioritise right donors	Donor Twin propensity scoring
Anticipate availability and willingness	Drift detection + graph fragility analysis
Encourage continued participation	Bridge Pairing emotional loop
Long-term engagement mechanisms	Karma system + constellation view
Feedback loops that feel human	AI-generated impact narratives
Work across diverse populations	Federated architecture + Voice IVR
Adapt to language and infrastructure	Bhashini stack + multi-channel rails
Efficiency while expanding regions	Resilience graph + Trust Fabric

Thank you.